

A virtual private network (VPN) service is a proxy server marketed to help users bypass Internet censorship such as geo-blocking and users who want to protect their communications against user profiling or MitM attacks on hostile networks.

A wide variety of entities provide VPN services for several purposes. Depending on the provider and the application, they do not always create a true private network. Instead, many providers simply provide an Internet proxy that uses VPN technologies such as OpenVPN or WireGuard. Commercial VPN services are often used by those wishing to disguise or obfuscate their physical location or IP address, typically as a means to evade Internet censorship or geo-blocking.

Providers often market VPN services as privacy-enhancing, citing security features, such as encryption, from the underlying VPN technology. However, when the transmitted content is not encrypted before entering the proxy, that content is visible at the receiving endpoint regardless of whether the VPN tunnel itself is encrypted for the inter-node transport. On the client side, configurations intended to use VPN services as proxies are not conventional VPN configurations. However, they do typically utilize the operating system's VPN interfaces to capture the user's data to send to the proxy. This includes virtual network adapters on computer OSes and specialized "VPN" interfaces on mobile operating systems. A less common alternative is to provide a SOCKS proxy interface.

In 2025, 1.75 billion people used VPNs. By 2027, this market has been projected to grow to \$76 billion. As of 2022, recommendation websites for VPNs tended to be affiliated with or even owned by VPN service providers, and VPN service providers often make misleading claims on their products.

Reasons for use

While research on user behavior and motivations for using VPNs is relatively limited compared to technical literature, studies generally find that users are motivated by concerns over

security and privacy, particularly protection against hackers. In contrast, a study of 349 college students found that students were more likely to use VPNs to access entertainment content than for privacy-related reasons. Another study of 90 technically savvy users reported that those motivated by privacy concerns, rather than by practical needs such as accessing geo-blocked content, were more likely to continue using VPNs over time. Surveys have also found that users tend to distrust free VPN services and express concern about providers collecting or selling their data.

VPN usage has also been observed to increase in response to content restrictions, social media taxes, and the implementation of age verification laws.

Accessing geo-restricted content

VPNs allow users to bypass regional restrictions by hiding their IP address from the destination server and simulating a connection from another country.

Improving privacy on public Wi-Fi

Where public Wi-Fi networks do not provide isolated encryption for each connected device, VPN services can provide a certain level of protection. When in use, potential eavesdroppers on the network can only observe that a connection to the VPN server is made by a user's device. As of June 2025, however, approximately 98% of human-generated internet traffic was encrypted using TLS through the HTTPS protocol; when TLS is used, network eavesdropping can only point out the IP addresses or hostnames a user is connecting to. Interception of network requests by a bad actor in the form of a Man-in-the-middle attack will most likely result in a certificate warning in being displayed in the user's browser.

SSL stripping, the practice of downgrading a connection to unencrypted HTTP, doesn't always result in a browser warning, although this has been partly mitigated by the implementation of HTTP Strict Transport Security.

Improving privacy

Activists and journalists working in restrictive or authoritarian regions can use VPNs to help maintain anonymity and protect sensitive communications.

By geographical region

As of 2025, four of the top six countries of VPN adoption rates from 2020 through the first half of 2025 were in the Middle East: UAE #1, Qatar #2, Oman #5 and Saudi Arabia #6. Aside from bypassing a block of content, it is thought that bypassing restriction of voice over internet protocol (VoIP) services, like WhatsApp, Skype, and FaceTime are motivating factors.

Criticism and limitations

Research has generally found that non-specialist users often have flawed mental models of VPNs and misunderstand the extent of the protection they provide. Such misconceptions may persist even among active VPN users.

Most users discover VPN services through review websites, which can be influenced by commercial incentives, with some relying on paid reviews and auctioning off the top review spot.

Users are commonly exposed to misinformation on the VPN services market, which makes it difficult for them to discern fact from false claims in advertisements. According to research by Consumer Reports, 12 out of 16 surveyed service providers had poor privacy and security practices and also made hyperbolic claims. The New York Times has advised users to reconsider whether a VPN service is worth their money. VPN services are not sufficient for protection against browser fingerprinting.

The provider may log the user's traffic, although this depends on the individual company. Users can still be tracked through tracking cookies even if the user's IP address is hidden.

A VPN service is not in itself a means for good Internet privacy. The burden of trust is simply transferred from the Internet service provider to the VPN service provider.

Legality

China

In China, unlawful use of VPNs may result in criminal prosecution under the relatively obscure Supreme People's Court guidelines: the Criminal Information Technology System Security Offense Adjudicative Guidelines and the Damage to Telecommunications Market Integrity Adjudicative Guidelines.

According to the guidelines, however, the simple use of typical VPN tunnels is not inherently unlawful because it does not achieve the elements of a computer crime, i.e. intrusion or unlawful control of a computer. VPN providers themselves can be prosecuted because providing a type of VPN in a way that severely disrupts the telecommunications market constitutes the offense of unlawful business operations. Additionally, if a VPN is used to commit illegal activities, then its provision could fall under aiding and abetting a crime. This was the logic applied by Chinese police in the widely publicized case involving a Chinese programmer who was penalized on grounds he used an unapproved international connection to provide internet consulting services to a company for 1,058,000 RMB in unlawful income.

Russia

Russia banned various VPN service providers in 2021. Law No. 276-FZ (2017) requires VPN/anonymizer services to prevent access to sites on the government blacklist; it prohibits owners of virtual private network (VPN) services and internet anonymizers from providing access to websites banned in Russia. The obligation is codified via amendments adding Article 15.8 to the Information Law and enforced by Roskomnadzor.

North Korea

Communication through other countries' communication networks without approval within North Korea is subject to a blanket criminal ban. The 2023 revision of the Radio Wave Control Law provides penalties including fines and "up to three months of unpaid labor or punishment by labor education".

Iran

VPNs are subject to general criminalization, but with discretion by the government to allow certain permissible uses. Use of filtering-circumvention tools (e.g., VPN services) is prohibited unless legally authorized by permit under the Supreme Council of Cyberspace's 2024 resolution (cl. 6).

Comparison of commercial virtual private network services

Privacy

In 2018 PC Magazine recommended that users consider choosing a provider based in a country with no data retention laws because that makes it easier for the service to keep a promise of no logging. PC Magazine and TechRadar also suggested that users read the provider's logging policy before signing up for the service, because some providers collect information about their customers' VPN usage.

Technical features

Notes

Encryption

Notes

Definitions

The following definitions clarify the meaning of some of the column headers in the comparison tables above.

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